
MATH 80601A *Bayesian modelling*

Practice midterm examination

Exam booklet

Examiner: Léo Belzile,

Instructions: The time allotted for the examination is 180 minutes. You may answer in either English or French. No written material may be brought into the examination.

There are a total of 21 marks available in the exam paper, the distribution of which can be found in the right margin in brackets.

You must hand back the **exam booklet** at the end of the examination.

Crib sheet: A Poisson random variable with mean $\lambda > 0$ denoted $\text{Poisson}(\lambda)$, has density

$$f(x) = \frac{\lambda^x}{x!} \exp(-\lambda), \quad x \in 0, 1, \dots$$

A Gaussian random variable with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$, denoted $\text{Gauss}(\mu, \sigma^2)$, has density

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{1}{2\sigma^2}(x - \mu)^2\right\}, \quad x \in \mathbb{R}.$$

An inverse gamma random variable with shape $\alpha > 0$ and rate $\beta > 0$, denoted $\text{inv.gamma}(\alpha, \beta)$, has density

$$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{-\alpha-1} \exp(-\beta/x), \quad x \in \mathbb{R}.$$

where $\Gamma(\cdot)$ is the gamma function. The latter satisfies $\Gamma(a+1) = a\Gamma(a)$ for $a \geq 0$.

Question:	1	2	Total
Points:	5	16	21
Score:			

Question 1. Short answer questions

- 1.1 If X is a continuous random variable with density function $f(x)$, is $f(x_0)$ a probability for given x_0 ? Justify your answer. [1 pt]

Solution: The integral of the density over the domain is unity, but the only constraint otherwise is positivity. Counterexample is $\text{expo}(\lambda)$, with density $\lambda \exp(-\lambda x)$, which reduces to $\lambda > 0$ at zero.

- 1.2 **Inversion sampling:** for an absolutely continuous distribution function F with quantile function F^{-1} and $U \sim \text{unif}(0, 1)$ uniform, prove that $F^{-1}(U)$ has distribution function F . [2 pts]

Solution: The distribution function $F(X) \sim \text{unif}(0, 1)$, so applying the monotone transformation F^{-1} on either sides gives $F^{-1}\{F(X)\} = X = F^{-1}(U)$, since the quantile function is the inverse of F .

- 1.3 Give an example of a distribution $p(Y | \theta)$ which may admit as conjugate prior for θ [2 pts]
 i. the gamma distribution,
 ii. the beta distribution.

Solution: Poisson distribution with gamma prior, binomial distribution with beta prior. For example, if $Y \sim \text{Poisson}(\theta)$ and $\theta \sim \text{gamma}(a, b)$, then $p(y | \theta) \propto \theta^y \exp(-\theta)$ and $p(\theta) \propto \theta^{a-1} \exp(-\theta b)$, then $p(\theta | y) \propto \theta^{y+a-1} \exp\{-\theta(1+b)\}$ which is $\text{gamma}(y+a, 1+b)$.

Question 2. Modelling unemployment rates

We consider a times series of $n = 142$ monthly harmonised unemployment rates (in percentage) of women in Norway for the period running from April 2000 to January 2012. We treat the sample $\mathbf{y}_{1:n} = (y_1, \dots, y_n)^\top$ as the realization of an autoregressive process of order 1, denoted AR(1), of the form

$$Y_t | Y_{t-1} = y_{t-1} \sim \text{Gauss}(\alpha + \phi y_{t-1}, \nu), \quad t = 1, \dots, n; \quad (1)$$

where α is a location parameter, ν is the conditional **variance** and ϕ is the autoregressive parameter. Throughout, we perform inference conditional on the observed initial value y_0 for March 2000, treating the latter as a fixed quantity. Recall that an AR(1) process is stationary if $|\phi| < 1$, with the unconditional variance of $\sigma^2/(1 - \phi^2)$.

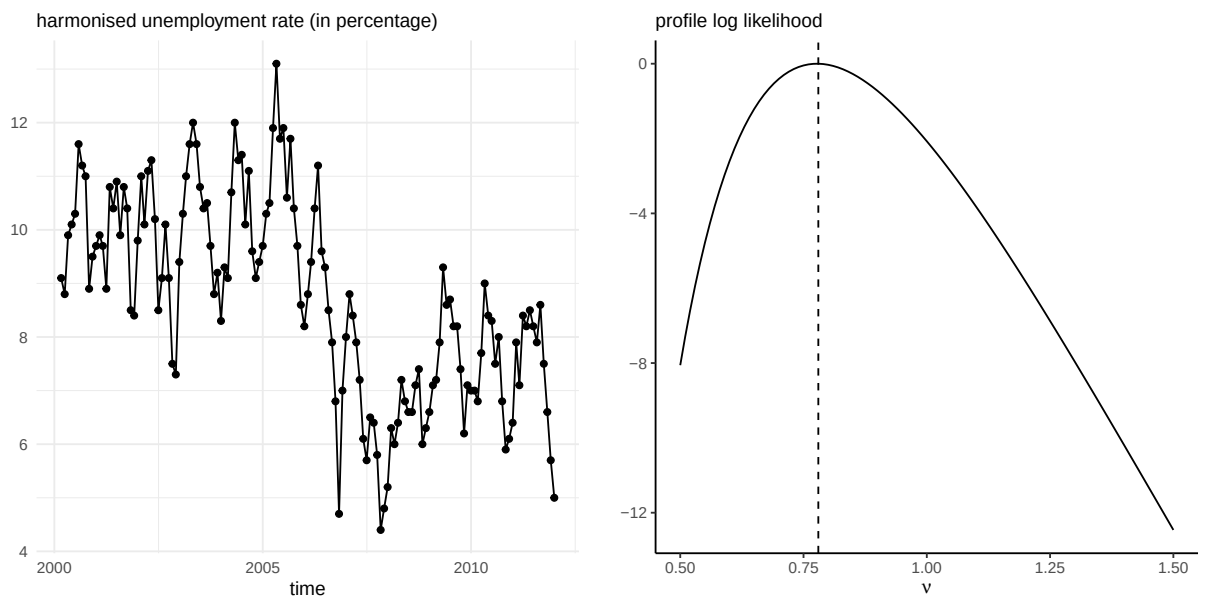


Figure 1: Left: time series of monthly women harmonised unemployment rate over time. Right: Profile log likelihood for the variance parameter v of eq. (1). The curve has been shifted down so that the profile log likelihood is zero at the maximum likelihood estimate. The vertical dashed line indicates the maximum likelihood estimate.

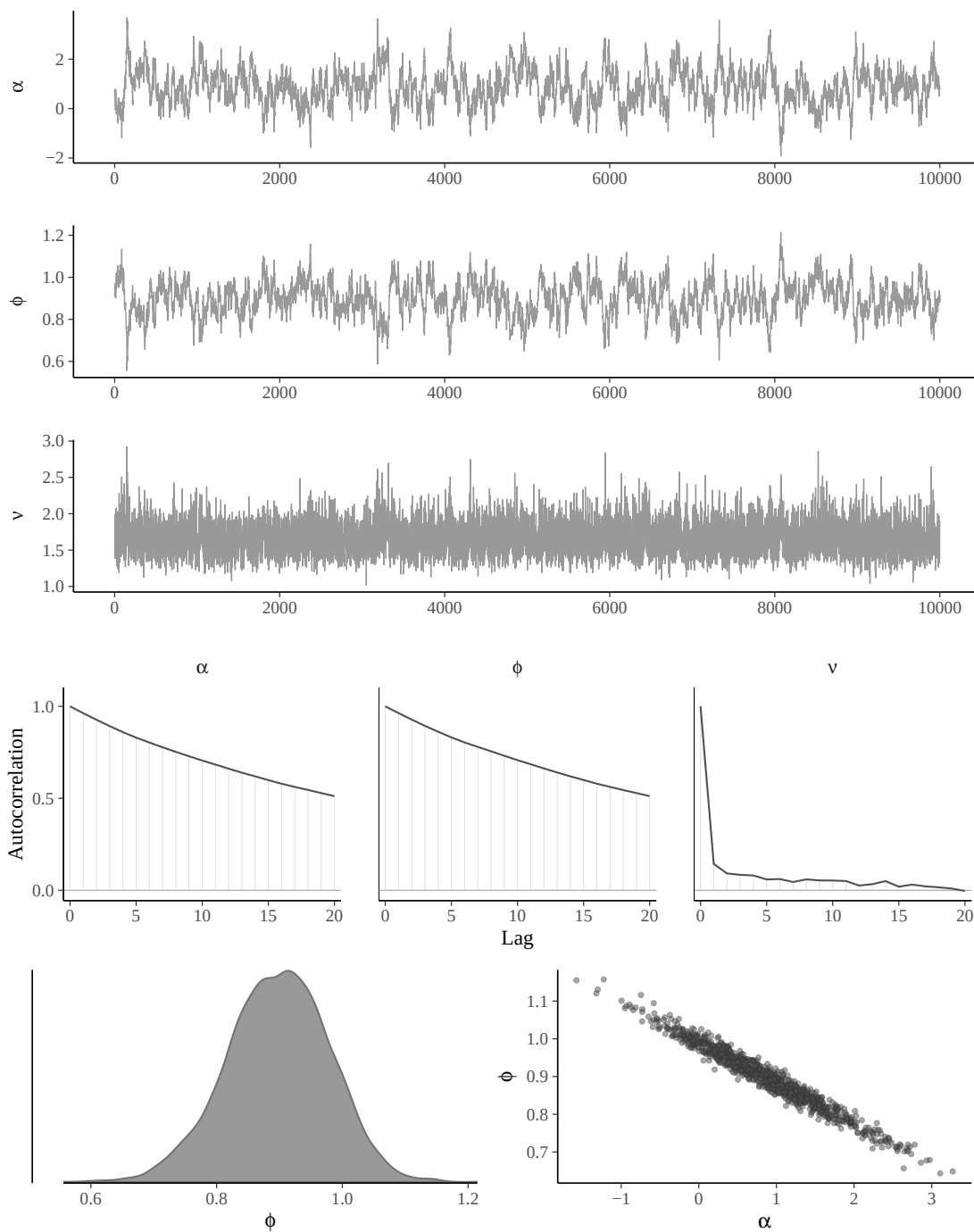


Figure 2: Diagnostics of the Markov chains obtained by running a Gibbs sampler to obtain 10 000 posterior draws from α, ϕ, v , with trace plots (top), correlograms of individual chains (middle), histogram of the autocorrelation coefficient (bottom left) and scatterplot of pairs (α, ϕ) (bottom right). The estimated effective sample sizes are $ESS(\alpha) = 226$, $ESS(\phi) = 217$ and $ESS(v) = 3644$.

2.1 Write down the joint likelihood $L(\phi, \alpha, \nu; \mathbf{y}_{1:n})$.

[2 pts]

Solution: The joint density can be expressed as the product of conditional densities, using Markov property

$$\begin{aligned} p(\mathbf{y}_{1:n} | \alpha, \phi, \nu) &= \prod_{t=1}^n p(y_t | y_{t-1}, \alpha, \phi, \nu) \\ &= (2\pi\nu)^{-n/2} \exp \left[-\frac{1}{2\nu} \sum_{t=1}^n \{y_t - \alpha - \phi y_{t-1}\}^2 \right] \end{aligned}$$

and treating this expression as a function of (α, ϕ, ν) gives the likelihood $L(\phi, \nu; \mathbf{y}_{1:n})$.

2.2 Consider the reference prior

[1 pt]

$$p(\alpha, \phi, \nu) \propto \nu^{-1} \mathbf{I}(\nu > 0) \mathbf{I}(\alpha \in \mathbb{R}) \mathbf{I}(\phi \in \mathbb{R}). \quad (2)$$

Is this prior proper? Justify your answer.

Solution: Since $\nu > 0$, the prior is improper because the integral of x^{-1} over the positive half line diverges. It is also improper because $\int_{\mathbb{R}} 1 = \infty$.

2.3 Using the reference prior of Equation (2), derive and provide explicit expressions for the parameters of the posterior distributions

[4 pts]

$$p(\alpha | \phi, \nu, \mathbf{y}_{1:n})$$

$$p(\phi | \alpha, \nu, \mathbf{y}_{1:n})$$

$$p(\nu | \alpha, \phi, \mathbf{y}_{1:n})$$

Explain how you would use these to obtain posterior draws using a Gibbs sampler.

Solution: Straightforward calculations show that

$$\begin{aligned} p(\alpha | \phi, \nu, \mathbf{y}_{1:n}) &\sim \text{Gauss} \left\{ \frac{\sum_{t=1}^n (y_t - \phi y_{t-1})}{n}, \frac{\nu}{n} \right\} \\ p(\phi | \alpha, \nu, \mathbf{y}_{1:n}) &\sim \text{Gauss} \left\{ \frac{\sum_{t=1}^n (y_t - \alpha) y_{t-1}}{\sum_{t=1}^n y_{t-1}^2}, \frac{\nu}{\sum_{t=1}^n y_{t-1}^2} \right\} \\ p(\nu | \alpha, \phi, \mathbf{y}_{1:n}) &\sim \text{inv.gamma} \left\{ \frac{n}{2}, \frac{1}{2} \sum_{t=1}^n (y_t - \alpha - \phi y_{t-1})^2 \right\} \end{aligned}$$

Gibbs sampler: starting from say the maximum likelihood estimator, alternate sampling from each conditional distribution in turn, treating other parameters as fixed.

- 2.4 Suppose you to select a more informative prior for the variance by taking $v \sim \text{inv.gamma}(\beta, \beta)$. Derive the expected value of the $\text{inv.gamma}(\beta, \beta)$ prior and use moment matching to select the value β such that the apriori expected value for v is 1.5. [2 pts]

Solution: The density of the inverse gamma is

$$f(x; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{-\alpha-1} \exp(-\beta/x) \mathbb{I}(x > 0)$$

so we have for $\alpha > 1$

$$\begin{aligned} E(X) &= \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{-(\alpha-1)-1} \exp(-\beta/x) dx \\ &= \frac{\beta \Gamma(\alpha-1)}{\Gamma(\alpha)} \int_0^\infty f(x; \alpha-1, \beta) dx \end{aligned}$$

and noting that $\Gamma(x+1) = x\Gamma(x)$, we recover $E(X) = \beta/(\alpha-1)$. If we set $\beta/(\alpha-1) = 1.5$, then $\beta = 3$.

- 2.5 Figure 2 on page 3 shows diagnostic plots for the Gibbs sampler. Comment on the stationarity and mixing, and diagnose any problems related to sampling. [2 pts]

Solution: The chains all appear to have reached their stationary distribution. It's obvious that there is a mixing problem for the mean parameters, owing to the strong correlation between (α, ϕ) that leads to slow exploration of the state space and low effective sample sizes.

- 2.6 Figure 2 shows some posterior draws of ϕ exceed one, which corresponds to AR(1) nonstationary models with infinite variance. How could you enforce stationarity? [1 pt]

Solution: A simple option is to take a truncated prior which has range $[-1, 1]$ or assigns little to no prior probability there, e.g., a penalized complexity prior. Another option is to shrink towards zero.

- 2.7 In line with the previous two subquestions, describe modifications of your algorithm that would improve the efficiency of the sampling of posterior draws. [2 pts]

Solution: This could be remedied by making block updates using Gibbs (this is a linear regression) or via a within Metropolis–Hastings step, or by reparametrizing the model using, e.g., the centered parametrization $\mu + \phi(y_{t-1} - \mu)$.

With a simple redefinition $p(\phi) = \frac{1}{2}I(|\phi| \leq 1)$, the conditional $p(\phi \mid \alpha, \nu, \mathbf{y}_{1:n})$ for the Gibbs sampler is simply a truncated Gaussian, which can be sampled using either accept-reject or inversion sampling. Otherwise, one would need to use Metropolis-within-Gibbs for ϕ .

- 2.8 For February 2012, the analysis yields a one-step ahead forecast, obtained from the posterior predictive distribution $p(\tilde{y}_{n+1} \mid \mathbf{y}_{1:n})$, of 5.39% with an equitailed 50% credible interval of [4.48%, 6.30%]. Explain in your own words how you would calculate such interval based on the posterior draws $(\alpha_{(b)}, \phi_{(b)}, \sigma_{(b)})$ ($b = 1, \dots, B$). [2 pts]

Solution: For each of the B draws of the Markov chain, say $(\alpha_{(b)}, \phi_{(b)}, \sigma_{(b)})$ ($b = 1, \dots, B$) sample a single realization from the posterior predictive $\text{Gauss}(\alpha_{(b)} + \phi_{(b)}y_n, \sigma_{(b)}^2)$.

Compute the empirical quartiles and return these.